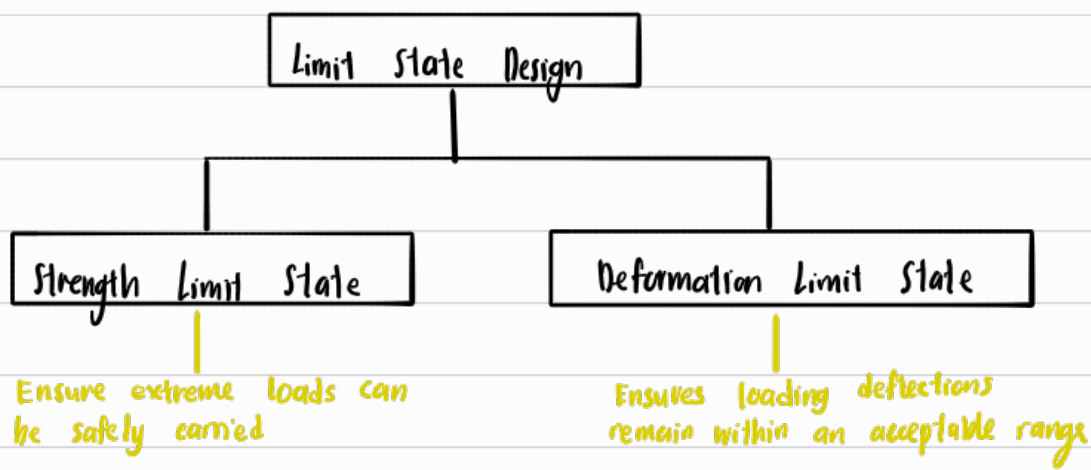




Factor of Safety, SF

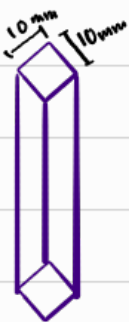
$$[SF]_{\text{strength}} = \frac{\text{Load or stress capacity}}{\text{Actual load or stress}}$$

$$[SF]_{\text{deformation}} = \frac{\text{Deflection or deformation capacity}}{\text{Actual deflection or deformation}}$$

If $SF > 1 \Rightarrow \text{Safe}$

$SF < 1 \Rightarrow \text{Failure}$

Example 1:



If yield stress, $\sigma_y = 250 \text{ MPa}$, what is the max load it can carry without failing? Assume a SF of 1.25.

$$SF = \frac{\text{capacity}}{\text{Actual}}$$

$$1.25 = \frac{250 \text{ MPa}}{\text{Actual}}$$

$$\text{Actual} = 200 \text{ MPa}$$

$$\sigma_{\text{max of SF}} = 1.25 = \frac{F_{\text{max}}}{A}$$

$$200 \text{ MPa} = \frac{F_{\text{max}}}{(10 \text{ mm})^2}$$

$$F_{\text{max}} = 20 \text{ kN} \#$$

Example 2:

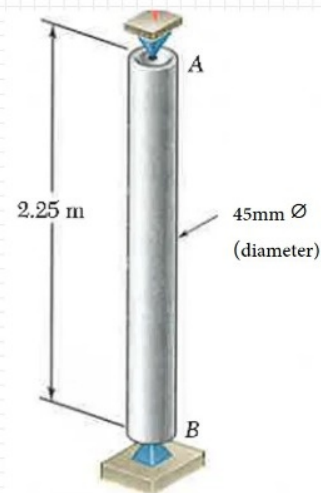
A 2.25m long solid steel circular column of diameter 45mm experiences an axial tension force of 300kN.

a) What is the strength limit state factor of safety of this system?

b) The column is not to change in length by more than 0.1%.

Is this requirement met? What is the deflection limit state factor of safety here?

Assume $E_{\text{steel}} = 200,000 \text{ MPa}$ & Yield strength = 250 MPa



$$a) \sigma_{\text{Actual}} = \frac{300 \times 10^3}{\pi \left(\frac{45}{2} \times 10^{-3} \right)^2} = 188.6 \text{ MPa}$$

$$SF_{\text{Strength}} = \frac{250}{188.6} = 1.33 *$$

$$\begin{aligned} b) \text{ Young Modulus, } E &= \frac{\sigma}{\epsilon} = \frac{\sigma L}{\Delta L} \\ \Delta L_{\text{Actual}} &= \frac{\sigma L}{E} \\ &= \frac{(188.6)(2.25)}{200,000} \\ &= 2.122 \times 10^{-3} \text{ m} \\ &= 2.122 \text{ mm} \end{aligned}$$

$$\text{When } \% \Delta L = 0.1\%$$

$$\Delta L_{\text{Capacity}} = 2.25 \left(\frac{0.1}{100} \right) = 2.25 \times 10^{-3} \text{ m} > 2.122 \times 10^{-3} \text{ [Requirement met]}$$

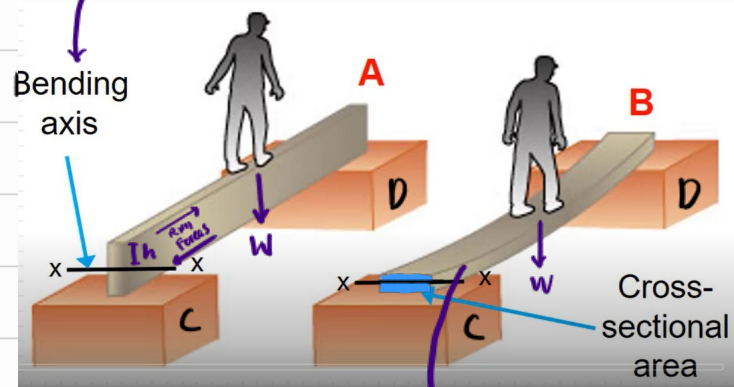
$$SF_{\text{deflection}} = \frac{2.25 \times 10^{-3}}{2.122 \times 10^{-3}} = 1.06$$

1st moment of Area [Bending Inertia]

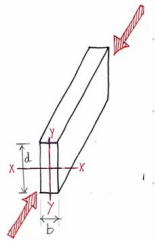
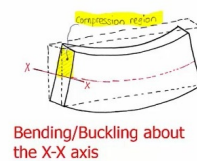
- Measure of a shape's resistance to bending [Bending stiffness]

- When we apply an external torque to an object, the fibers of the object shortened (compressed) or lengthened (tension). The reaction forces of the fibers will be in the opposite direction of the external normal stress, producing a internal torque to counteract the external torque
- Higher the internal torque, higher the bending inertia

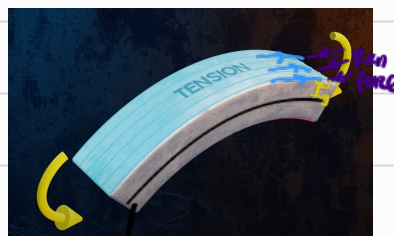
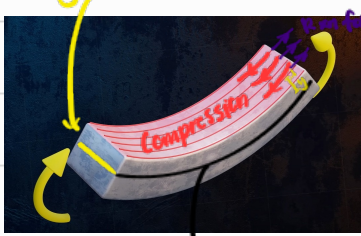
Same as the rotational axis [If Block C or D was not there, the member will rotate about the x-axis]



- A section has two possible axes about which it might buckle – the 'x' and 'y' axes. These axes are 90° apart.
- The bending/buckle axes divides the region of the beam in compression from the region in tension.



Bending Axis



the axis with no stress [Neutral Axis]

Neutral Axis

$$\text{Bending Strain, } \epsilon = \frac{\Delta L}{L} = \frac{CD - AB}{AB} = \frac{(R+y)\theta - R\theta}{R\theta} = \frac{y}{R}$$

$$\epsilon = \frac{\sigma}{E} \Rightarrow \text{Bending stress, } \sigma_B = \frac{E y}{R}$$

$$\begin{aligned} \text{Bending Moment, } M_B &= \int \sigma_B y dA \\ &= \int \left(\frac{E y}{R} \right) y dA \\ &= \frac{E}{R} \int y^2 dA \end{aligned}$$

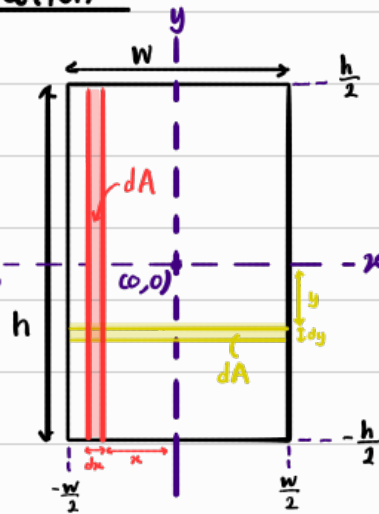
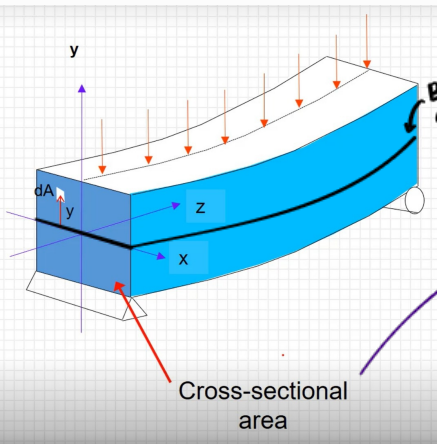
$$M_B = \frac{E}{R} (I) \quad \text{Bending Inertia [2nd moment of area]}$$



* Only need this for Week 7

$$\therefore I_B = \int y^2 dA$$

I For a rectangular section



$\therefore w \sim \text{width}$
 $h \sim \text{height}$

Assuming x-axis is the bending axis

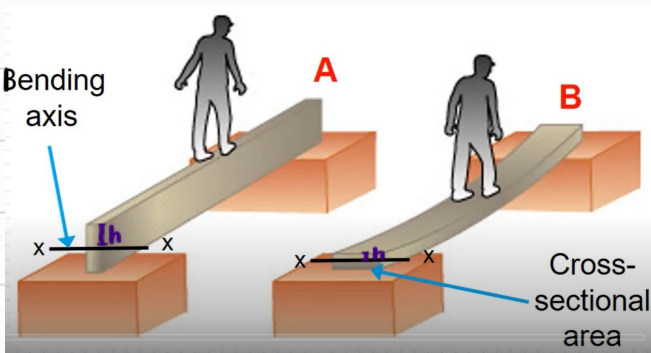
$$\begin{aligned} I_x &= \int y^2 dA \\ &= \int_{-h/2}^{h/2} (y^2 w) dy \\ &= \left[\frac{w y^3}{3} \right]_{-h/2}^{h/2} \end{aligned}$$

$$I_x = \frac{w h^3}{12}$$

Assuming y-axis is the bending axis

$$\begin{aligned} I_y &= \int x^2 dA \\ &= \int_{-w/2}^{w/2} (x^2 h) dx \\ &= \left[\frac{h x^3}{3} \right]_{-w/2}^{w/2} \end{aligned}$$

$$I_y = \frac{h w^3}{12}$$



Looking back at this diagram, since the bending axis from the front view is the x-axis

$$\Rightarrow I_x = \frac{w h^3}{12}$$

$$\therefore h_A > h_B, I_{x_A} > I_{x_B}$$

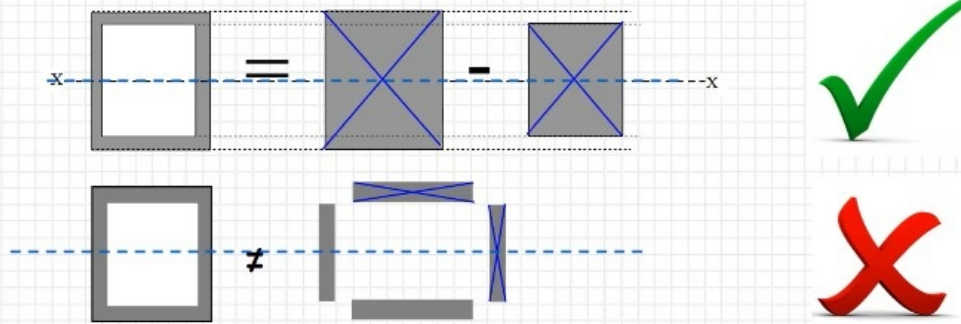
$\therefore$ That's why A is much harder to bend about the x-axis.

I for circular section

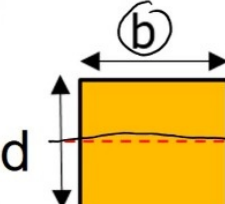
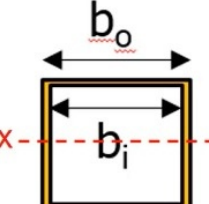
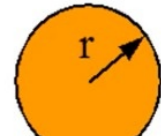
$$I = \frac{\pi r^4}{4} \quad [\text{complex derivation}]$$

I for Symmetrical Cross-sections made up of Rectangles & circles

HOWEVER the centroid of all the parts MUST lie on the same centroidal axis line.



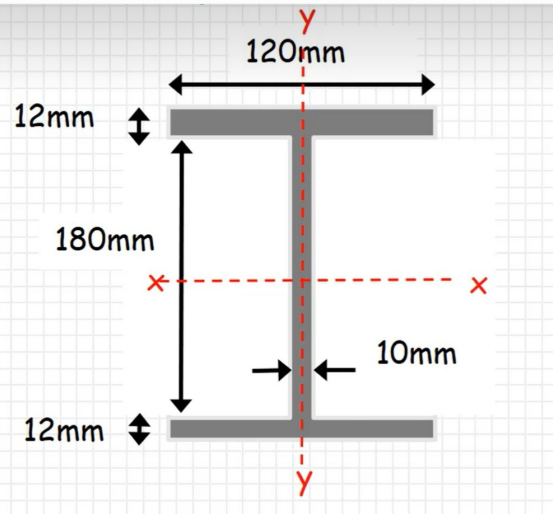
Example :


$$I_{xx} = \frac{bd^3}{12}$$

$$I_{xx} = \frac{b_o d_o^3}{12} - \frac{b_i d_i^3}{12}$$

$$I = \frac{\pi r^4}{4}$$

$$I = \frac{\pi(r_o^4 - r_i^4)}{4}$$

Example 2:

Find I for the universal beam below.



About x-axis

$$\Sigma I_x = \frac{120 (204)^3}{12} - 1 \left[\frac{55 (180)^3}{12} \right]$$

$$= 31.4 \times 10^6 \text{ mm}^4 \quad \#$$

About y-axis

$$\Sigma I_y = \frac{12 (120)^3}{12} + \frac{180 (10)^3}{12} + \frac{12 (120)^3}{12}$$

$$\Sigma I_y = 3.47 \times 10^6 \text{ mm}^4 \quad \#$$

Buckling

- Sudden deformation under increasing compression

Critical buckle load/buckle strength

$$P_{cr} = \frac{\pi^2 EI}{L^2}$$

Example 1:

A 2.25m long pin-ended solid steel circular column diameter 45mm is now loaded in **compression**.

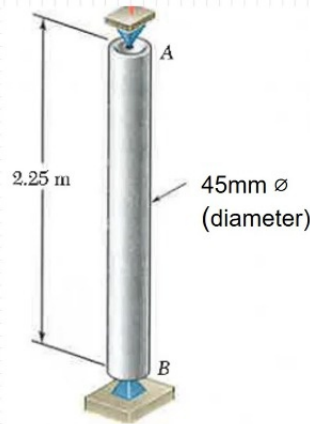
a) What is its buckle strength?

b) What is its buckle strength if it is twice as long?

Assume $E_{\text{steel}} = 200,000 \text{ MPa}$, Yield strength = 250 MPa

& $I = \pi r^4 / 4 = 2.01 \times 10^5 \text{ mm}^4$

$$P_{cr} = \frac{\pi^2 EI}{L^2}$$



$$\text{a) } P_{cr} = \frac{\pi^2 (200,000) (2.01 \times 10^{-7})}{2.25^2} = 78,372 \text{ N} = 78.4 \text{ kN} \#$$

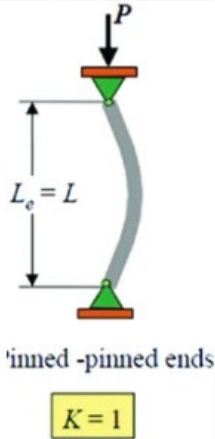
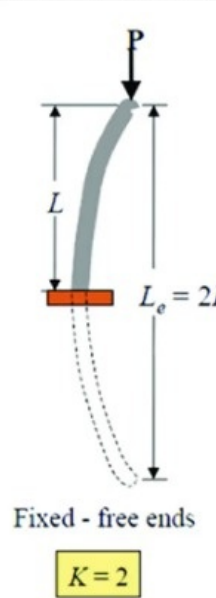
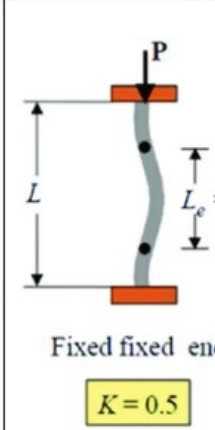
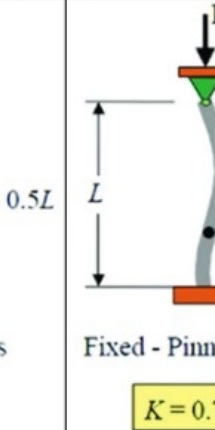
$$\text{b) } P_{cr} = \frac{\pi^2 (200,000) (2.01 \times 10^{-7})}{(2.25 \times 2)^2} = 19,593 \text{ N} = 19.5 \text{ kN} \#$$

Buckle formula for columns with different end restraints

$$P_{cr} = \frac{\pi^2 EI}{(kL)^2} = \frac{\pi^2 EI}{L_e^2}$$

L_e = the **effective length** of the member = kL where k = the effective length factor

L_e = the distance between points of zero moment or the distance over which you get a $\frac{1}{2}$ -sine curve

 inched - pinned ends $K = 1$	 Fixed - free ends $K = 2$	 Fixed fixed ends $K = 0.5$	 Fixed - Pinned ends $K = 0.7$
$P_{cr} = \frac{\pi^2 EI}{L^2}$	$P_{cr} = \frac{\pi^2 EI}{4L^2}$	$P_{cr} = \frac{4\pi^2 EI}{L^2}$	$P_{cr} = \frac{2.045\pi^2 EI}{L^2}$

Example 1:

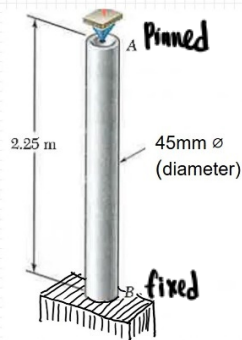
A 2.25m long solid steel circular column of diameter 45mm is loaded in compression. End B of the column is now rigidly connected to a support i.e. it is not free to rotate.

What is the column buckle strength?

Assume $E_{\text{steel}} = 200,000 \text{ MPa}$, Yield strength = 250 MPa

& $I_{\text{section}} = 2.01 \times 10^5 \text{ mm}^4$

$$P_{cr} = \frac{\pi^2 EI}{(L_e)^2}$$



Since its Pinned & Fixed supported, $K = 0.7 \Rightarrow L_e = 0.7 (2.25) = 1.575$

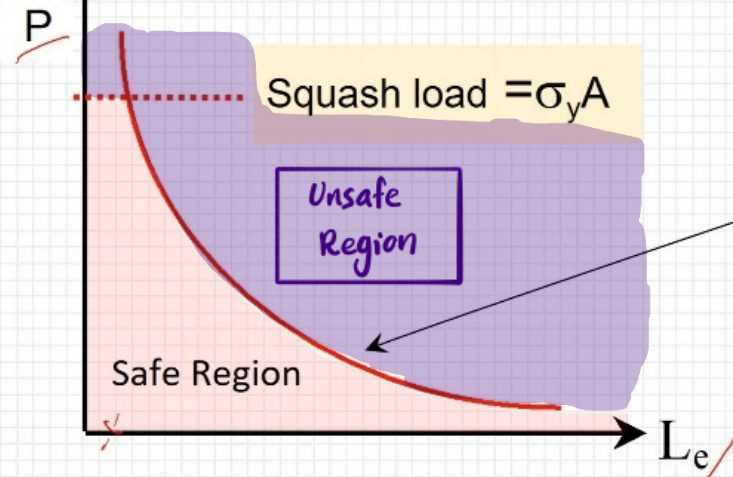
$$P_{cr} = \frac{\pi^2 (200,000) (2.01 \times 10^{-7})}{(1.575)^2} = 159,942 \text{ N}$$

$$= 160 \text{ kN}$$

Axial Compression Failure Envelope

Axial Compression Load

The most a section can carry in compression is its squash load

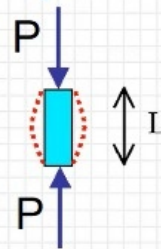


$$P_{cr} = \frac{\pi^2 EI}{(L_e)^2}$$

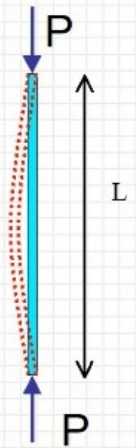
$$f(x) = \frac{k}{x^2} \quad \text{where } k = \pi^2 EI$$

$$L_e \rightarrow \infty, P_{cr} \rightarrow 0$$

$$L_e \rightarrow 0, P_{cr} \rightarrow \infty \quad \text{This is not physically possible.}$$



Long members fail by buckling
Short members fail by yielding

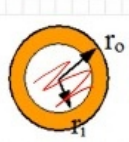


Example

A pin ended 20mmx1.5mm thick steel circular hollow section is loaded in compression.

Assume $E=200,000\text{MPa}$ and Yield strength = 250MPa

Plot the compression failure envelope for this member and note where the following two cases lie on this envelope and thus what mode of failure is for each.



$$I = \frac{\pi(r_o^4 - r_i^4)}{4} = \frac{\pi(10^4 - 8.5^4)}{4} = 3754\text{mm}^4$$

$$A = \pi(r_o^2 - r_i^2) = \pi(10^2 - 8.5^2) = 87\text{mm}^2$$

a) $L_e = 0.5\text{m}$

Point lies on buckle curve above Squash load

→ Failure mode = yield

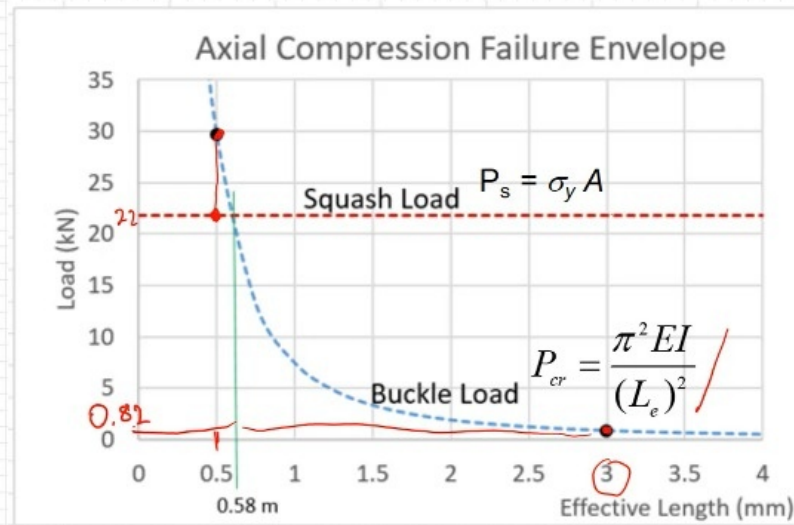
Failure load = 22 kN

b) $L_e = 3\text{m}$

Point lies on buckle curve below Squash load

→ Failure mode = buckling

Failure load = 0.82 kN

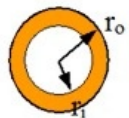


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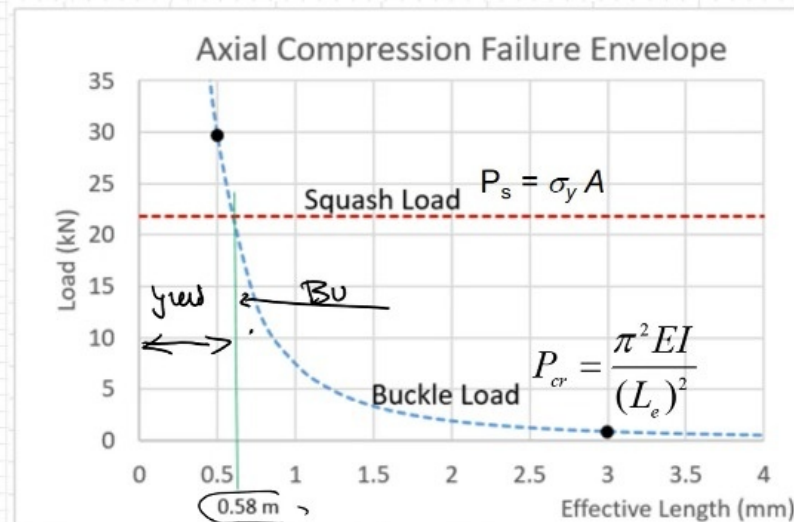
$$A = \pi(r_o^2 - r_i^2) = \pi(10^2 - 8.5^2) = 87\text{mm}^2$$

Length where failure mode changes from Yield to buckling

$$\sigma_y A = \frac{\pi^2 EI}{L_e^2}$$

$$250 \times 87 = \frac{\pi^2 \times 200,000 \times 3754}{(L_e)^2}$$

$$\rightarrow L_e = 583\text{mm} = 0.58\text{m}$$

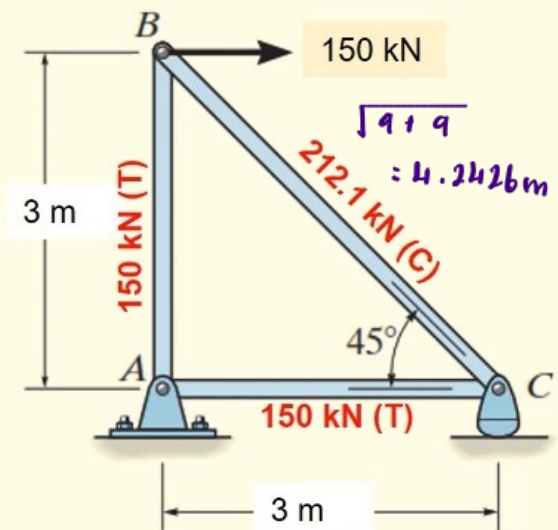


Example

A methods of Joints analysis of a truss structure yields the internal member force values shown.

- Choose the most weight efficient member sizes for all three members based on a strength limit state design. Choose sections from CHS (Circular hollow section) tables based on a minimum strength factor of safety of 2.
- Which member will cause limit state strength failure of the system if the load increases further, so what will be the overall failure mode of this system?
- Based on your chosen member sizes, calculate the true (unfactored) horizontal movement at B.

$$\sigma_y = 150 \text{ MPa} ; E = 100,000 \text{ MPa}$$



a) For member AB & AC [same load]

$$SF = \frac{\text{load capacity}}{\text{Actual load}}$$

$$F_{\text{capacity}} = 2(150) = 300 \text{ kN}$$

$$\therefore \sigma_{\text{yield}} = \frac{300,000}{\text{Area}}$$

$$\text{Area} = \frac{300,000}{150} = 1200 \text{ mm}^2 \text{ [Minimum Area needed to satisfy SF]}$$

$\therefore$ Since these members are under tension, it won't undergo buckling.

Thus we don't need to check for minimum 1st moment of Area, I

Nearest Area from graph is 1230 mm^2 , so choose that row

$$\begin{aligned} \therefore \text{Actual Factor of safety} &= \frac{\text{load capacity}}{\text{Actual load}} = \frac{\text{Area g'ten from table}}{(\sigma_{\text{yield}})(\text{Area})} \\ &= \frac{150}{(250 \text{ MPa})(1230 \text{ mm}^2)} \\ &= \frac{150 \text{ kN}}{150 \text{ kN}} \\ &= 2.05 \text{ [Just slightly greater than 1]} \end{aligned}$$

TABLE 1.1(a)
DIMENSIONS AND PROPERTIES
TUBELINE CIRCULAR HOLLOW SECTIONS
GRADE C250L0 (AS 1163)

DIMENSION AND RATIOS				PROPERTIES										PROPERTIES FOR DESIGN TO AS 4100	
Designation	Mean diameter mm	External surface area m ² /m	Weight kg/m	Section modulus mm ³	Area moment of inertia mm ⁴	Radius of gyration mm	Torsion constant mm ²	Torsion modulus mm ³	Form factor	Compactness ¹	Z _x	Z _y			
100.0 x 5.0 CHS	105.0	0.314	24.4	30.6	2710	8.65	100	130	56.5	17.3	209	1.00	30.6	C	130
125.0 x 5.0 CHS	130.0	0.519	39.3	33.0	2510	8.07	97.7	126	56.8	16.1	195	1.00	33.0	C	126
150.0 x 5.0 CHS	155.0	0.820	54.2	35.3	2200	6.14	89.7	97.4	47.3	10.3	147	1.00	35.3	C	97.4
175.0 x 5.0 CHS	180.0	1.025	69.1	37.7	2100	5.61	84.8	90.8	47.7	9.61	138	1.00	37.7	C	90.8
200.0 x 5.0 CHS	205.0	1.230	83.9	40.0	1850	4.75	68.0	64.1	38.5	5.49	96.1	1.00	40.0	C	64.1
225.0 x 5.0 CHS	230.0	1.435	98.8	42.4	1550	3.81	51.0	54.3	30.9	4.89	82.0	1.00	42.4	C	54.3
250.0 x 5.0 CHS	255.0	1.640	113.7	44.8	1300	3.14	40.0	46.7	24.2	3.55	69.9	1.00	44.8	C	46.7
275.0 x 5.0 CHS	280.0	1.845	128.6	47.2	1050	2.50	30.0	38.1	18.5	2.93	57.6	1.00	47.2	C	38.1
300.0 x 5.0 CHS	305.0	2.050	143.5	49.6	850	1.98	21.0	30.9	13.9	2.30	46.4	1.00	49.6	C	30.9
325.0 x 5.0 CHS	330.0	2.255	158.4	52.0	650	1.46	12.0	23.7	9.2	1.67	35.2	1.00	52.0	C	23.7
350.0 x 5.0 CHS	355.0	2.460	173.3	54.4	450	0.94	3.0	16.5	4.5	1.05	24.0	1.00	54.4	C	16.5
375.0 x 5.0 CHS	380.0	2.665	188.2	56.8	250	0.42	0.0	9.0	1.5	0.63	13.0	1.00	56.8	C	9.0
400.0 x 5.0 CHS	405.0	2.870	203.1	59.2	50	0.00	0.0	0.0	0.0	0.00	0.0	1.00	59.2	C	0.0

NOTES:
1. This table is calculated in accordance with AS 4100 using design yield stress $f_y = 250 \text{ MPa}$ and design tensile strength $f_t = 300 \text{ MPa}$ as per AS 4100 table 2.1 for AS 1163 grade C250L0.
2. Grade C250L0 is cold formed and therefore is allocated the CF residual stress classification in AS 4100.
3. C = Compact Section; N = Non-compact Section; S = Slender Section, as defined in AS 4100.

cont.

a) For member BC

$$SF = \frac{\text{load capacity}}{\text{Actual load}}$$

$$F_{\text{capacity}} = 2(212.1) = 424.2 \text{ kN}$$

$$\therefore \sigma_{\text{yield}} = \frac{424200}{\text{Area}}$$

$$\text{Area} = \frac{424200}{250} = 1696.8 \text{ mm}^2 \text{ [Minimum Area needed to satisfy SF]}$$

$\therefore$ Since this member is under compression, it will undergo buckling.

Thus we need to also check for minimum 2nd moment of Area, I.

$$F_{\text{Buckling}} = \frac{\pi^2 EI}{Le^2}$$

K = 1 beuz its pin-pin

$$424200 = \frac{\pi^2 (200,000) I}{(1)(4.2426)}$$

$$I = 3.87 \times 10^6 \text{ mm}^4 \text{ [Minimum I needed to satisfy SF]}$$

$\therefore$ This means we have to find a CHS w/ $A \geq 1696.8 \text{ mm}^2$ and $I \geq 3.87 \times 10^6 \text{ mm}^4$

$\rightarrow$ That will be the row of (139.7 x 5.0 CHS)

$$\therefore \text{Actual } SF_{\text{yield}} = \frac{\text{load capacity}}{\text{Actual load}}$$

$$= \frac{(\sigma_{\text{yield}})(\text{Area})}{212.1 \times 10^3}$$

$$= \frac{(250)(2120)}{212.1 \times 10^3} \text{ Area gotten from table}$$

$$= \frac{(250)(2120)}{212.1 \times 10^3}$$

$$= 2.499 \text{ [Just slightly greater than 1]}$$

$$\therefore \text{Actual } SF_{\text{buckling}} = \frac{\text{Buckling capacity}}{\text{Actual buckling}}$$

$$= \frac{(\pi^2)(200,000)(4.81 \times 10^6)}{(1)(4242.64)^2}$$

I value gotten from table

$$= \frac{(\pi^2)(200,000)(4.81 \times 10^6)}{(1)(4242.64)^2} \div 212.1$$

$$= 2.485 \text{ [Just slightly greater than 1]}$$

b) Since the minimum SF for AB and AC is 2.05, while the minimum SF for BC is 2.485

↳ Members AB & AC will fail 1st . #

↳ Failing by yield #

c) $E = \frac{FL}{A\Delta L}$

$\Delta L = \frac{FL}{AE}$

$$\Delta L_{AB} = \frac{(150,000)(3000)}{(1230)(200000)}$$

$$= 1.829 \text{ mm}$$

$$\therefore L_{fAB} = 3000 + 1.829$$

$$= 3001.829 \text{ mm} \#$$

$$\therefore L_{fAC} = L_{fAB} = 3001.829 \text{ mm} \#$$

$$\Delta L_{BC} = \frac{(212,100)(4242.64)}{2120 \times 200000}$$

$$= 2.1223 \text{ mm}$$

$$\therefore L_{fBC} = 4242.64 + 2.1223$$

$$= 4240.52 \text{ mm} \#$$



Using cosine rule on new position:

$$4240.52^2 = 3001.829^2 + 3001.829^2 - 2(3001.829)(3001.829) \cos \theta$$

$$\theta = 89.873^\circ$$

$$\tan \alpha = \frac{x}{3000}$$

$$\tan(90 - \theta) = \frac{x}{3000}$$

$$x = 6.65 \text{ mm} \#$$